

D846

General Physics I Lab

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A lab group using a brand-new set of steel ramps wants to study how ramp angle affects how fast a cart reaches the bottom. The group's cart is painted bright yellow. Which statement is a testable hypothesis?

- A. How does ramp angle affect the cart?
- B. A yellow cart will move faster than other colors.
- C. If the ramp angle increases, then the time for the cart to reach the bottom will decrease.
- D. Ramps are interesting to study.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

In a free-fall lab done in a stairwell with good lighting, a student measures g as 9.52 m/s^2 . The accepted value is 9.80 m/s^2 . The student's timer is accurate to 0.01 s . What is the percent error?

- A. 0.28%
- B. 97.1%
- C. 2.9%
- D. -2.9%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A student's percent error for g is 5.0% , and her measured value is below the accepted value of 9.80 m/s^2 . What was her measured value?

- A. 4.90 m/s^2
- B. 9.31 m/s^2
- C. 10.29 m/s^2
- D. 9.75 m/s^2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A group applies different net forces to a cart on a low-friction track and plots force (N) on the y-axis against acceleration (m/s^2) on the x-axis. The best-fit line has a slope of 2.5. The track was borrowed from a nearby college. What does the slope represent?

- A. Friction, because the track came from a college.
- B. The cart's acceleration, 2.5 m/s^2 .
- C. The inverse of mass, 0.4 kg.
- D. The cart's mass, 2.5 kg.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Another group plots acceleration (m/s^2) on the y-axis against force (N) on the x-axis and gets a slope of 0.40. What is the cart's mass?

- A. Cannot be found from a graph.
- B. 2.5 kg
- C. 0.40 kg
- D. 4.0 kg

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A student using a laser rangefinder measures a distance of 12.0 m and a time of 3.2 s. The calculator shows 3.75 m/s. The rangefinder cost \$300. How should the speed be reported?

- A. 3.8 m/s
- B. 3.750 m/s, since the rangefinder is precise.
- C. 3.75 m/s
- D. 4 m/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A student adds lengths of 12.51 m and 3.2 m. How should the sum be reported?

- A. 15.7 m
- B. 15.710 m
- C. 16 m
- D. 15.71 m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

In a lab with new digital scales, every mass a group measures reads 0.20 g higher than the true value because the scale was not zeroed. The group plans to run 20 more trials. The lab runs on Tuesdays. What should they do?

- A. Run 20 more trials to average out the error.
- B. Buy newer digital scales.
- C. Zero the scale and correct the data; more trials will not remove a systematic error.
- D. Accept the data since all errors are random.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A group stretches a spring with masses from 0.1 kg to 0.5 kg and finds extension is proportional to force. The spring is from a well-known supply company. They conclude the spring will stretch proportionally with a 50 kg mass. What is wrong?

- A. The data support the relationship only from 0.1 to 0.5 kg; the spring may exceed its elastic limit.
- B. Nothing; the brand guarantees quality.
- C. Nothing; linear data always continue forever.
- D. They should have tested only one mass.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A 2.0 kg cart, recently repainted, is released from rest at a height of 0.50 m on a ramp and reaches a speed of 2.8 m/s at the bottom. The ramp is 1.5 m long. Using $g = 9.8 \text{ m/s}^2$, how much energy was transferred to thermal energy?

- A. 7.84 J
- B. About 2.0 J
- C. 0 J; energy was destroyed.
- D. 9.8 J

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

If the same 2.0 kg cart were released from 0.50 m on a frictionless ramp, what speed would it reach at the bottom? ($g = 9.8 \text{ m/s}^2$)

- A. 2.2 m/s
- B. 2.8 m/s
- C. 9.8 m/s
- D. 3.1 m/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A student writing a lab report with a clever title places this sentence in the Results section: Our data show that friction caused energy loss, supporting our hypothesis. The report is typed in a 12-point font. Where does this sentence belong?

- A. In the Results, where all findings go.
- B. In the title.
- C. In the Materials section.
- D. In the Discussion, because it interprets data and evaluates the hypothesis.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A lab group ~~using a brand new set of steel ramps~~ wants to study how ramp angle affects how fast a cart reaches the bottom. ~~The group's cart is painted bright yellow.~~ Which statement is a testable hypothesis?

Target word: testable hypothesis **Decides:** Measurable variables with a predicted relationship.

A	Adjacent category	A research question, not a hypothesis.
B	Camouflage	Color is not the variable under study.
C	Correct	Names variables and a falsifiable prediction.
D	Dropped element	No variables or prediction.

2. Answer C Concept 2

In a free-fall lab ~~done in a stairwell with good lighting~~, a student measures g as 9.52 m/s^2 . The accepted value is 9.80 m/s^2 . ~~The student's timer is accurate to 0.01 s .~~ What is the percent error?

Target word: compute percent error **Decides:** $|9.52 - 9.80| / 9.80 \times 100$.

A	Wrong stem number	The difference, not divided by accepted value.
B	Inverted operation	Experimental / accepted, not the error.
C	Correct	$0.28 / 9.80 \times 100 \approx 2.9\%$.
D	Dropped element	Drops the absolute value.

3. Answer B Concept 2 · transfer

A student's percent error for g is 5.0%, and her measured value is below the accepted value of 9.80 m/s^2 . What was her measured value?

Target word: solve for the measurement **Decides:** $9.80 \times (1 - 0.050)$.

A	Confusion set	Treats 5.0% as dividing by 2.
B	Correct	$9.80 - 0.49 = 9.31$.
C	Inverted operation	Adds the error when the value was below.
D	Wrong stem number	Subtracts 0.05, not 5% of 9.80.

4. Answer D Concept 3

A group applies different net forces to a cart on a low-friction track and plots force (N) on the y-axis against acceleration (m/s^2) on the x-axis. The best-fit line has a slope of 2.5. ~~The track was borrowed from a nearby college.~~ What does the slope represent?

Target word: meaning of slope **Decides:** $F = ma$, so $\Delta F / \Delta a = m$.

A	Camouflage	Where the track came from is irrelevant.
B	Confusion set	Acceleration is on the x-axis.
C	Inverted operation	That would be the slope of a vs. F .
D	Correct	Slope of F vs. a equals mass.

5. Answer B Concept 3 • transfer

Another group plots acceleration (m/s^2) on the y-axis against force (N) on the x-axis and gets a slope of 0.40. What is the cart's mass?

Target word: axes swapped **Decides:** Slope = $1/m$.

A	Absolute language	The slope gives it directly.
B	Correct	$m = 1 / 0.40 = 2.5 \text{ kg}$.
C	Inverted operation	Reads slope as mass without inverting.
D	Wrong stem number	Misreads 0.40 as 4.0.

6. Answer A Concept 4

A student ~~using a laser rangefinder~~ measures a distance of 12.0 m and a time of 3.2 s. The calculator shows 3.75 m/s. ~~The rangefinder cost \$300.~~ How should the speed be reported?

Target word: sig figs for division **Decides:** 3.2 has two significant figures.

A	Correct	Division keeps the fewest significant figures: two.
B	Camouflage	Timer precision limits the result.
C	Dropped element	Keeps calculator digits.
D	Confusion set	Rounds to one figure, fewer than the data justify.

7. Answer A Concept 4 • transfer

A student adds lengths of 12.51 m and 3.2 m. How should the sum be reported?

Target word: sig figs for addition **Decides:** Fewest decimal places: one.

A	Correct	Addition keeps one decimal place.
B	Inverted operation	Adds digits instead of rounding, creating false precision.
C	Confusion set	Applies the multiplication rule of two significant figures.
D	Dropped element	Keeps two decimal places.

8. Answer C Concept 5

In a lab ~~with new digital scales~~, every mass a group measures reads 0.20 g higher than the true value because the scale was not zeroed. The group plans to run 20 more trials. ~~The lab runs on Tuesdays.~~ What should they do?

Target word: fix the error type **Decides:** A consistent offset is systematic.

A	Inverted operation	Averaging reduces random, not systematic, error.
B	Camouflage	The scales are new; they were not zeroed.
C	Correct	Calibration fixes systematic error.
D	Absolute language	Not all errors are random.

9. Answer A Concept 6

A group stretches a spring with masses from 0.1 kg to 0.5 kg and finds extension is proportional to force. ~~The spring is from a well known supply company.~~ They conclude the spring will stretch proportionally with a 50 kg mass. What is wrong?

Target word: limit of conclusion **Decides:** 50 kg is far outside the tested range.

A	Correct	Extrapolation beyond range is not supported.
B	Camouflage	Brand does not extend the tested range.
C	Absolute language	Materials have limits.
D	Inverted operation	Fewer data make it worse.

10. Answer B Concept 7

A 2.0 kg cart, ~~recently repainted,~~ is released from rest at a height of 0.50 m on a ramp and reaches a speed of 2.8 m/s at the bottom. ~~The ramp is 1.5 m long.~~ Using $g = 9.8 \text{ m/s}^2$, how much energy was transferred to thermal energy?

Target word: energy transferred **Decides:** $PE = 9.8 \text{ J}$; $KE = 7.84 \text{ J}$.

A	Confusion set	The final kinetic energy.
B	Correct	$9.8 \text{ J} - 7.84 \text{ J} = 1.96 \text{ J} \approx 2.0 \text{ J}$, transferred to thermal energy by friction.
C	Absolute language	Energy is conserved; it was transferred.
D	Dropped element	The starting PE, not the difference.

11. Answer D Concept 7 · transfer

If the same 2.0 kg cart were released from 0.50 m on a frictionless ramp, what speed would it reach at the bottom? ($g = 9.8 \text{ m/s}^2$)

Target word: solve for speed **Decides:** $v = \sqrt{2gh}$.

A	Inverted operation	$\sqrt{gh}$, dropping the factor 2.
B	Wrong stem number	The speed with friction.
C	Dropped element	Forgets the square root.
D	Correct	$\sqrt{2 \times 9.8 \times 0.50} = \sqrt{9.8} \approx 3.1 \text{ m/s}$.

12. Answer D Concept 8

A student writing a lab report ~~with a clever title~~ places this sentence in the Results section: Our data show that friction caused energy loss, supporting our hypothesis. ~~The report is typed in a 12 point font.~~ Where does this sentence belong?

Target word: correct section **Decides:** It interprets data.

A	Confusion set	Results report data without interpretation.
B	Camouflage	Irrelevant.
C	Adjacent category	Materials list equipment.
D	Correct	Interpretation belongs in the discussion.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	2	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Testable hypotheses	A hypothesis is a testable, falsifiable statement that predicts how the independent variable will affect the dependent variable.	Writing a question, opinion, or untestable belief as a hypothesis.	An if...then statement naming measurable variables.
2	Percent error	Percent error = $ \text{experimental} - \text{accepted} / \text{accepted} \times 100\%$.	Dividing by the experimental value or dropping the absolute value.	An experimental value and an accepted value.
3	Meaning of a graph's slope	The slope of a linear graph is $\Delta y/\Delta x$ and carries physical meaning; for F vs. a it equals mass.	Reading the slope without regard to which variable is on which axis.	A best-fit line and a question about a physical quantity.
4	Significant figures	Multiply or divide: keep the fewest significant figures; add or subtract: keep the fewest decimal places.	Reporting every digit from the calculator.	A calculated result from measured values.
5	Systematic vs. random error	Random error scatters results and is reduced by repeated trials; systematic error shifts all results the same way and is fixed by calibration.	Believing more trials fix a zeroing or calibration error.	All results off in the same direction.
6	Limits of a conclusion	A conclusion is supported only within the range and conditions tested; extrapolation is a hypothesis to test.	Claiming a relationship holds far outside the tested range.	A conclusion reaches beyond the data collected.
7	Energy accounting in a system	In a closed system total energy is conserved; mechanical energy that seems lost has usually been transferred to thermal energy through friction.	Concluding energy was destroyed.	Final kinetic energy is less than starting potential energy.
8	Results vs. discussion in a lab report	The results section reports data, tables, and graphs; the discussion interprets them, explains errors, and states whether the hypothesis is supported.	Putting interpretation in results or leaving data out of results.	Deciding which section a sentence belongs in.

Course scope (public description): In this lab that follows the General Physics course, learners will develop the ability to conduct scientific investigations to answer questions using experimentation in the field of physics. The section emphasizes the application of the scientific method to solve problems, analyze data from experiments, and draw conclusions. Students will also learn to accurately summarize their findings and perform tasks with attention to detail. Throughout the lessons, students will engage in activities designed to enhance their scientific reasoning and written communication skills, ensuring they can effectively explain their results. They will practice identifying interconnections within systems. The course provides a foundational understanding of experimental techniques and data analysis, preparing learners to conduct independent investigations and apply their knowledge in real-world contexts.